

Capítulo 11

Chapter Precap

- Discutimos por que los modelos bayesianos multinivel son adecuados para el análisis de modelos con un gran número de parámetros, y algunas estrategias para interpretar estos modelos.
- Se presenta e interpreta un modelo con dos predictores cuantitativos, y una interacción entre ellos.
- Introducimos el análisis bayesiano de varianza (BANOVA) y los gráficos de ANOVA bayesiano y los utilizamos para investigar el modelo inicial presentado en el capítulo.
- Se utiliza un modelo de regresión logística multivariante y se interpreta mediante un ANOVA bayesiano.
- A continuación, explicamos cómo se pueden utilizar los modelos logísticos multivariados para interpretar la clasificación y el comportamiento de los oyentes.
- Por último, analizamos la selección de modelos y la especificación incorrecta, y algunos diagnósticos que pueden ayudar a encontrar y diagnosticar problemas en modelos grandes.

Problems with 'Large' Models

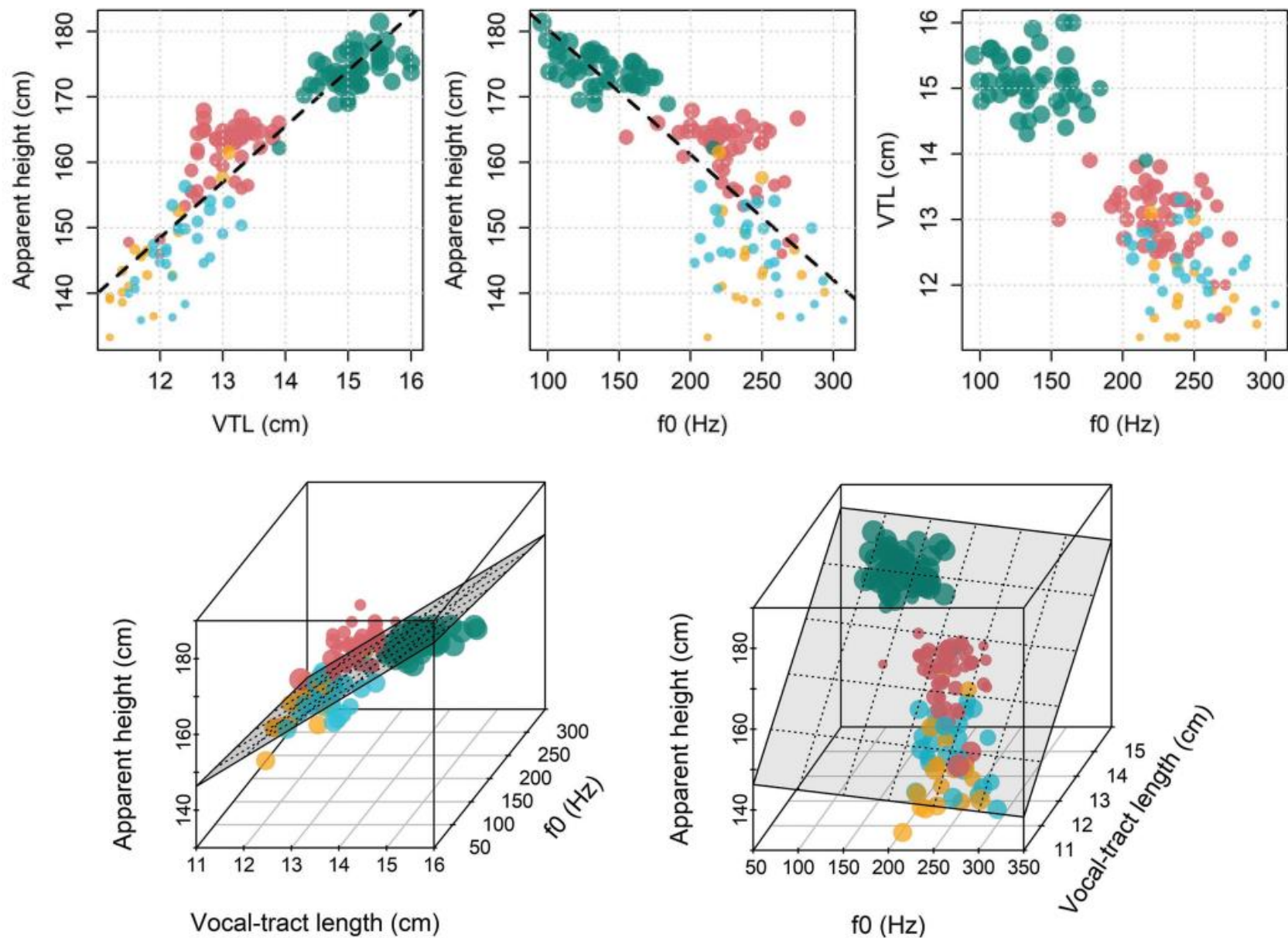
- 1 A model with many predictors may have 'too many' predictors. We will vaguely define 'extra' predictors as those which have no meaningful statistical association with your dependent variable, and which do not appreciably improve your model in any way. Sometimes, the 'extra' predictors have estimated values that are difficult to distinguish from those of the 'real' parameters, leading to incorrect conclusions.
- 2 A model with many parameters may have more difficulties with fitting/convergence, meaning you can't get good estimates for the model coefficients. Regardless of the approach to parameter estimation, more complicated models make it more difficult to find the optimal parameter values given the data and chosen model structure.
- 3 It can be difficult to interpret a model with hundreds (or thousands) of parameters. With so many parameters it is important to not miss the forest for the trees, that is, not consider every parameter in isolation to the detriment of understanding what information your fitted model provides in general.

Dealing with 'Large' Models

In order to avoid a situation where you end up with data you can't analyze or a model you can't interpret, it's worth considering the following questions *before* advancing to the data collection stage of your experiment:

- How will I analyze the data? Will I be able to carry out my planned analysis?
- What will the model structure be?
- What kind of results am I expecting?
- How will the expected results be reflected by the model parameters? How would different results manifest in the model parameters?

Multiple Quantitative Predictors



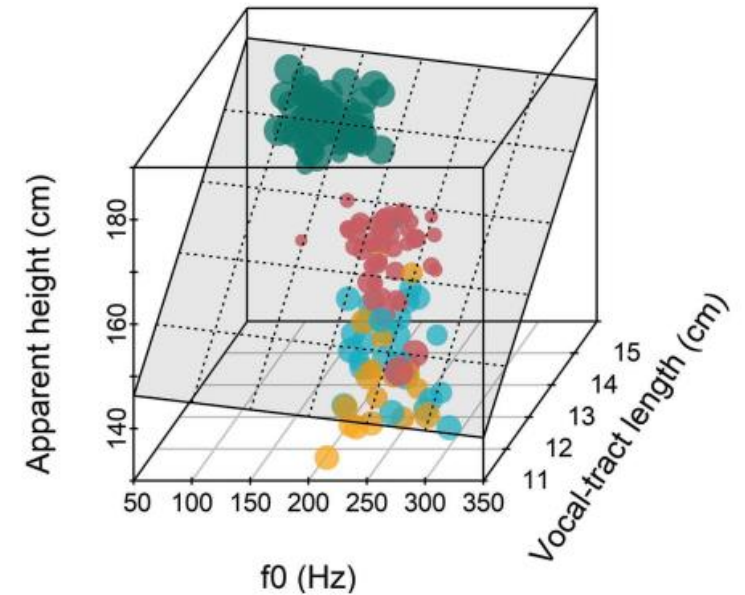
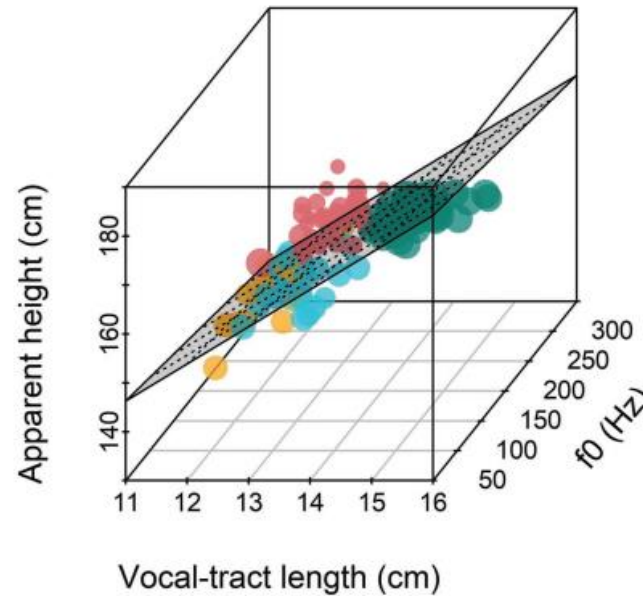
Multiple Quantitative Predictors

- Los planos tienen un intercepto y dos pendientes.

$$z = a + b \cdot x + c \cdot y$$

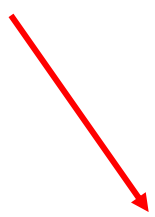
Diagram illustrating the multiple regression equation $z = a + b \cdot x + c \cdot y$. Red arrows point to the variables: z (vertical axis), a (intercept), b (coefficient for x), c (coefficient for y), x (horizontal axis), and y (depth axis).

La variable dependiente varia por el eje 'z' (vertical).



Interactions Between Quantitative Predictors

- Las 'interacciones' entre predictores cuantitativos se refieren al efecto sobre el producto de dichos predictores.

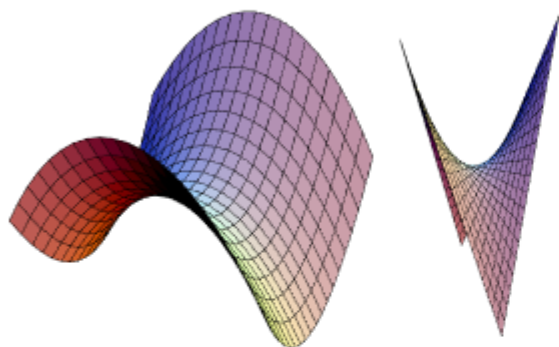

$$z = a + (b \cdot x) + (c \cdot y) + (d \cdot x \cdot y)$$

- Puede ingresarlos usando una fórmula de modelo ($y \sim x^*y$) o multiplicar sus predictores usted mismo, y usar esto como un predictor ($y \sim x + y + xy$, donde $xy = x \cdot y$).

Hyperbolic Paraboloid



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A hyperbolic paraboloid is the [quadratic](#) and [doubly ruled surface](#) given by the Cartesian equation

$$z = \frac{y^2}{b^2} - \frac{x^2}{a^2} \quad (1)$$

(left figure). An alternative form is

➡ $z = x y$ (2)

(right figure; Fischer 1986), which has [parametric equations](#)

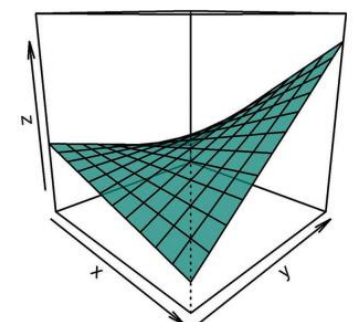
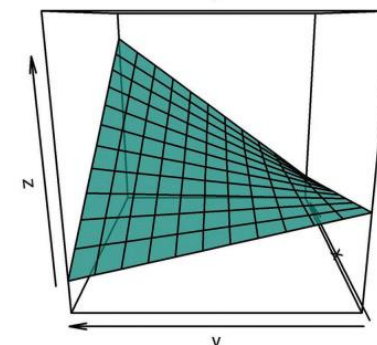
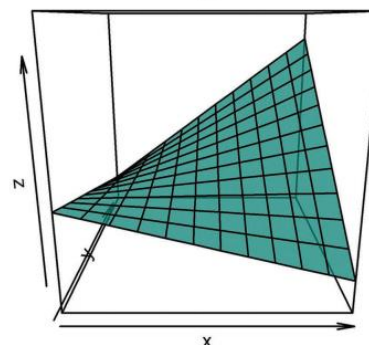
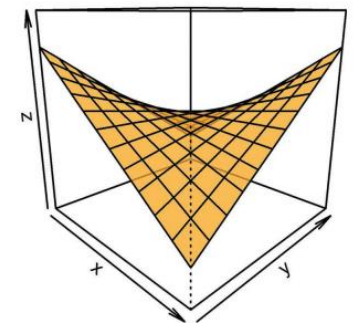
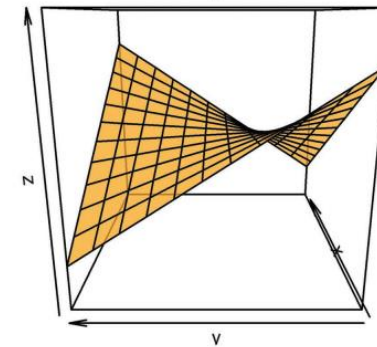
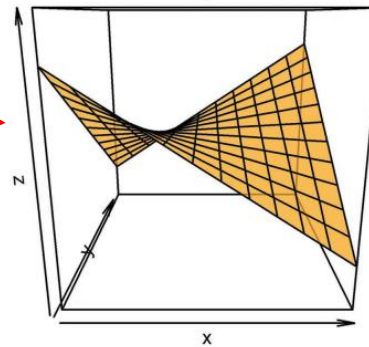
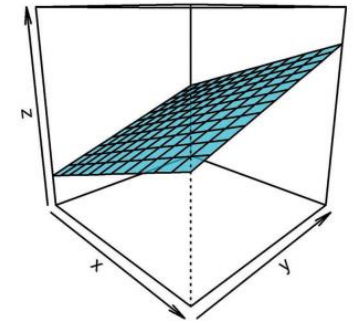
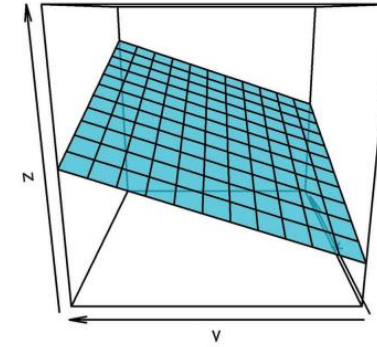
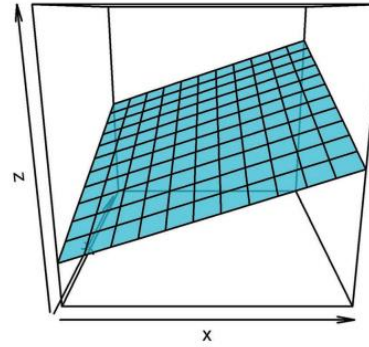
$$x(u, v) = u \quad (3)$$

$$y(u, v) = v \quad (4)$$

$$z(u, v) = u v \quad (5)$$

Interactions Between Quantitative Predictors

$$z = a + (b \cdot x) + (c \cdot y) + (d \cdot x \cdot y)$$



Linearly changing slopes

$$z = a + (b \cdot x) + \cancel{(c \cdot y)} + (d \cdot x \cdot y)$$

$$z = a + (b \cdot x) + (d \cdot x \cdot y)$$

$$z = a + (b) \cdot x + (d \cdot y) \cdot x$$

$$z = a + (b + d \cdot y) \cdot x$$

$$b = 2, d = 1, y = 1$$

$$z = a + (2 + 1 \cdot 1) \cdot x$$

$$z = a + 3 \cdot x$$

$$b = 2, d = 0$$

$$z = a + (b + d \cdot y) \cdot x$$

$$z = a + (2 + 0 \cdot y) \cdot x$$

$$z = a + 2 \cdot x$$

$$b = 2, d = 1, y = 5$$

$$z = a + (2 + 1 \cdot 5) \cdot x$$

$$z = a + 7 \cdot x$$

Center Your Predictors!

- ¡Tu coeficiente de pendiente x representa el efecto de x solo cuando $y = 0$!
- Cuando $y \neq 0$, el efecto de x puede ser mayor o menor.
- Al centrar los predictores, la pendiente de x es igual al efecto en la media de y .


$$b = 2, d = 0$$

$$z = a + (b + d \cdot y) \cdot x$$

$$z = a + (2 + 0 \cdot y) \cdot x$$

$$z = a + 2 \cdot x$$


$$b = 2, d = 1, y = 5$$

$$z = a + (2 + 1 \cdot 5) \cdot x$$

$$z = a + 7 \cdot x$$

Data and Research Questions

```
library (brms)
library (bmb)
options (contrasts = c('contr.sum', 'contr.sum'))

data (exp_data)

# center VTL
exp_data$vtl_original = exp_data$vtl
exp_data$vtl = exp_data$vtl - mean (exp_data$vtl)

# center and scale f0
exp_data$f0_original = exp_data$f0
exp_data$f0 = exp_data$f0 - mean(exp_data$f0)
exp_data$f0 = exp_data$f0 / 100
```

(Q1) What do we do with all these parameters? How do we know what to focus on, and where to begin?

Description of the Model

```
height ~ vtl*f0*A*G + (vtl*f0*A*G|L) + (1|S)
```

```
height ~ Intercept + vtl + f0 + A + G +  
          A:G1 + vtl:f0 + vtl:A + vtl:G + f0:A + f0:G +  
          vtl:f0:A + vtl:f0:G + vtl:A:G + f0:A:G + vtl:f0:A:G
```

```
a = Intercept + A          + G          + A:G  
b = vtl        + vtl:A      + vtl:G      + vtl:A:G  
c = f0         + f0:A        + f0:G        + f0:A:G  
d = vtl:f0     + vtl:f0:A   + vtl:f0:G   + vtl:f0:A:G
```

Description of the Model

```
height ~ vtl*f0*A*G + (vtl*f0*A*G|L) + (1|S)
```

$$\text{height}_{[i]} \sim t(\nu, \mu_{[i]}, \sigma)$$

$$\mu_{[i]} = a_{[i]} + (b_{[i]} \cdot \text{vtl}_{[i]}) + (c_{[i]} \cdot f0_{[i]}) + (d_{[i]} \cdot f0_{[i]} \cdot \text{vtl}_{[i]})$$

...

```
a = Intercept + A + G + A:G
b = vtl + vtl:A + vtl:G + vtl:A:G
c = f0 + f0:A + f0:G + f0:A:G
d = vtl:f0 + vtl:f0:A + vtl:f0:G + vtl:f0:A:G
```


Fitting the Model

```
# Fit the model yourself
```

```
priors = c(brms::set_prior("student_t(3,0, 12)", class = "Intercept"),  
           brms::set_prior("student_t(3,0, 12)", class = "b"),  
           brms::set_prior("student_t(3,0, 12)", class = "sd"),  
           brms::set_prior("lkj_corr_cholesky (2)", class = "cor"),  
           brms::set_prior("gamma(2, 0.1)", class = "nu"),  
           brms::set_prior("student_t(3,0, 12)", class = "sigma"))
```

```
model_height_vtl_f0 =
```

```
  brms::brm (height ~ vtl*f0*A*G + (vtl*f0*A*G|L) + (1|S), data = exp_data,  
            chains = 4, cores = 4, warmup = 1000, iter = 5000, thin = 4,  
            prior = priors, family = "student")
```

Advantages of Bayesian Multilevel Models for 'Large' Models

- Tres problemas con los modelos grandes son:

1. Es más probable que tenga "demasiados" predictores.

Los modelos multinivel ayudan con estos.

2. Más difícil encontrar la solución.

3. Más difícil de interpretar el modelo.

El ANOVA bayesiano ayuda con esto.

Comparison with lmer

- lmer es bastante similar a brms, incluso utilizando la agrupación parcial adaptativa para estimar efectos "aleatorios".
- Hay algunas diferencias importantes:
 - lmer no utiliza valores a priori para las estimaciones de parámetros.
 - Shrinkage solo se aplica a los efectos aleatorios, no a ningún otro parámetro (es decir, varianzas o correlaciones de efectos aleatorios).
 - Los intervalos solo se proporcionan para 'efectos fijos'.

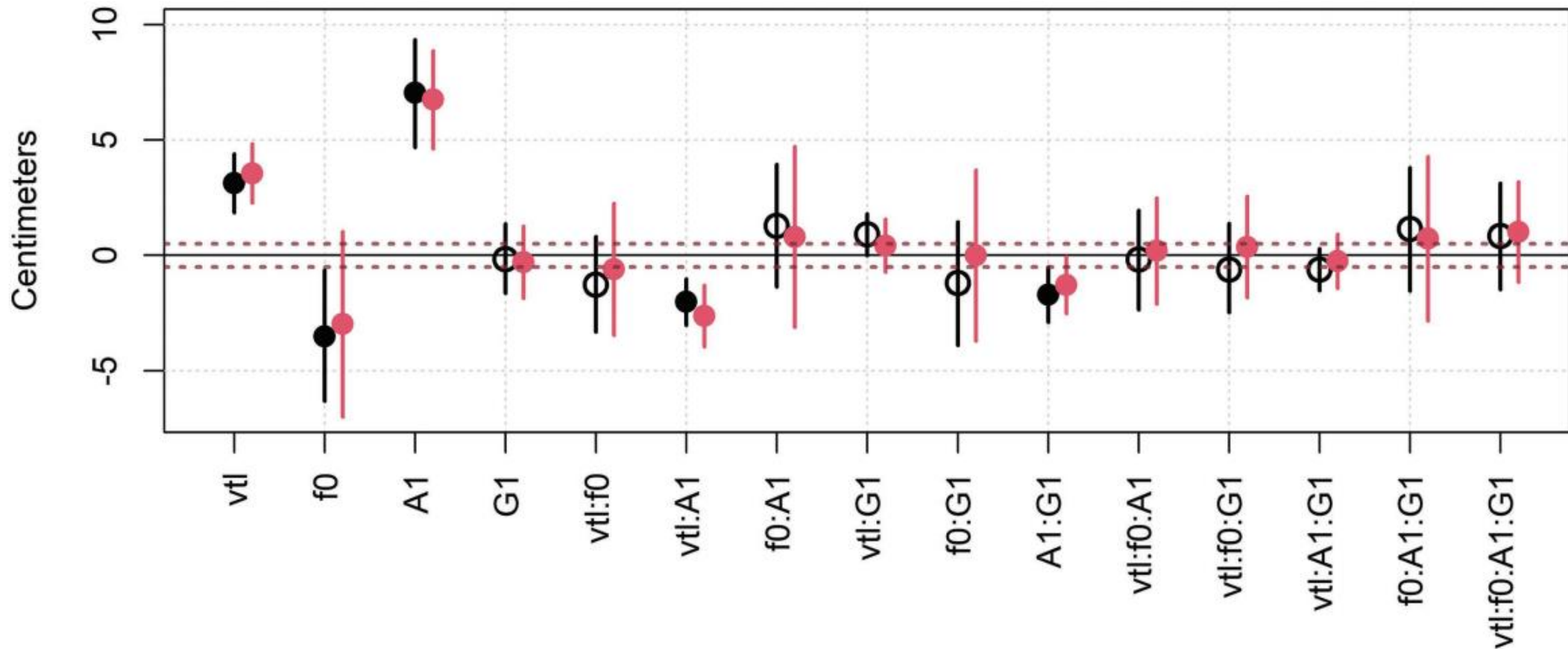
Comparison with lmer

```
# Fit the model yourself
lme_model_height_vt1_f0 =
  lme4::lmer (height ~ vt1*f0*A*G + (vt1*f0*A*G|L) + (1|S),
             data=exp_data, verbose = TRUE,
             control=lme4::lmerControl(optCtrl=list(maxfun= 20000),
             optimizer="bobyqa"))
```

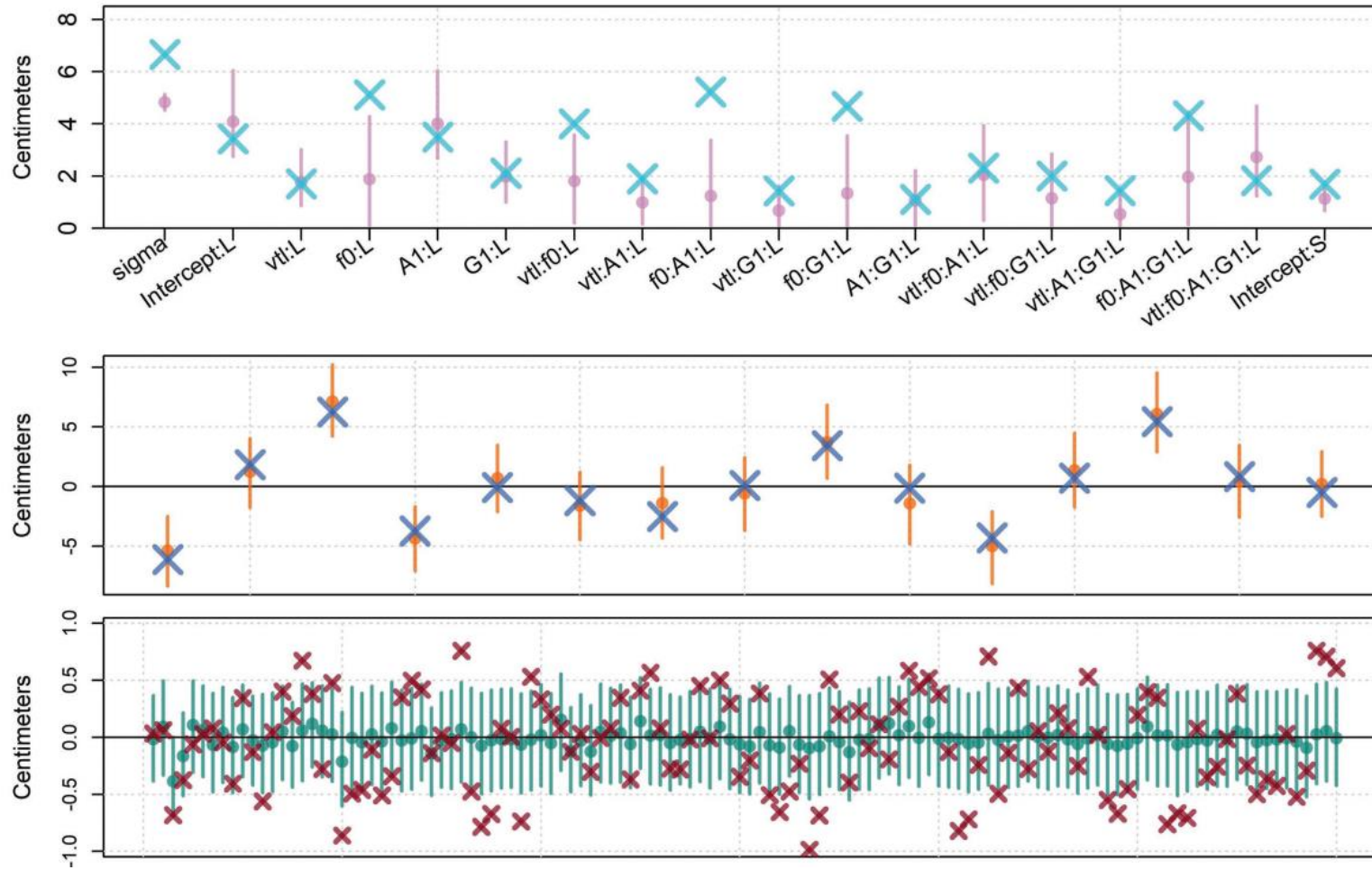
```
boundary (singular) fit: see help('isSingular')
```

Comparison: Fixed Effects

- Los efectos fijos son casi iguales entre brms (rojo) y lmer (negro).



Comparison: Random Effects and Correlations



Sin intervalos:

- Incertidumbre incierta.
- ¿Qué se diferencia realmente de cero?

ANalysis Of VAriance

- El ANOVA ve la variación en la variable dependiente como la suma de un conjunto de **componentes de variación** relacionados con los predictores de nuestro modelo. Por ejemplo:

```
height ~ C + (C|L) + (1|S)
```

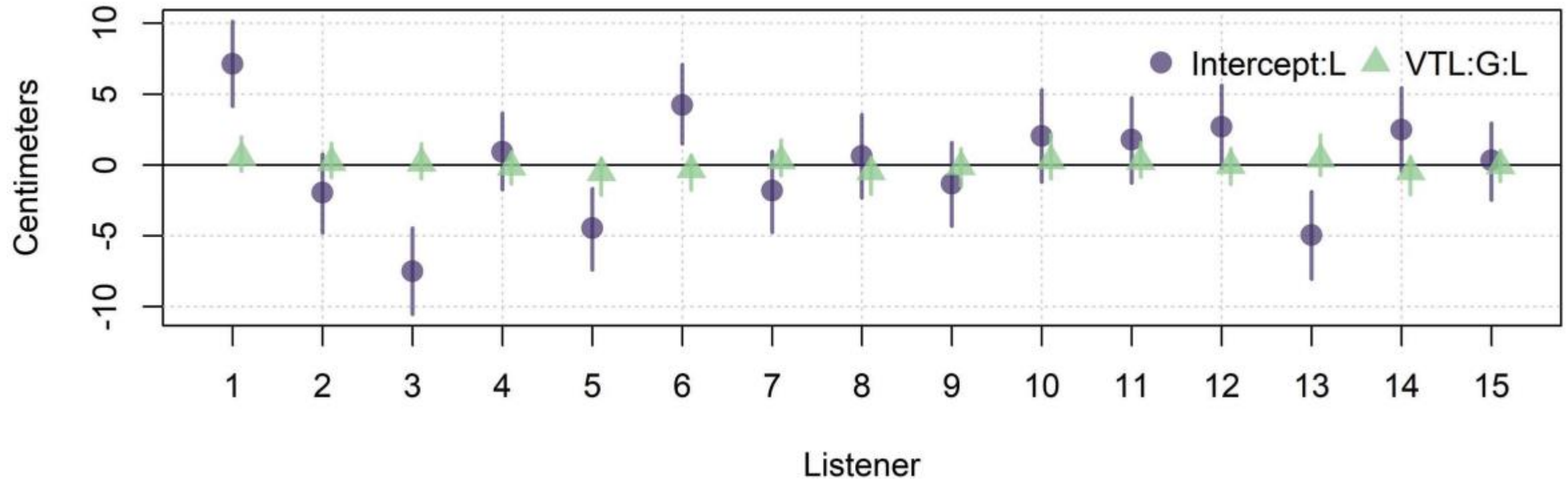
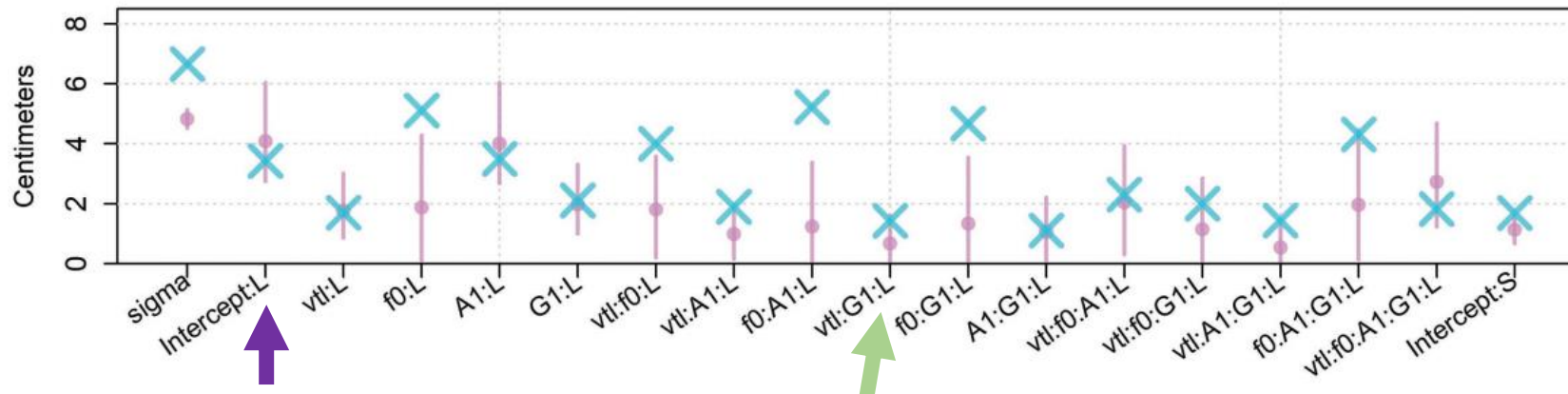
- 3 parameters representing the fixed effects for apparent speaker category C: C1, C2, C3.
- 15 parameters representing the listener-dependent intercepts.
- 45 (15 · 3) parameters representing the listener-dependent effects for parameters C1, C2, and C3.
- 139 parameters representing the speaker-dependent intercepts.

ANalysis Of VAriance

- Los lotes están vinculados semánticamente/temáticamente, no se trata solo de parámetros agrupados al azar.
- Cuando dividimos la variación en parámetros agrupados temáticamente, estamos haciendo una descomposición de la variación de la variable dependiente similar a un ANOVA.

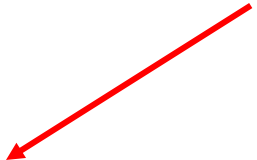
“When moving to multilevel modeling, the key idea we want to take from the analysis of variance is the estimation of the importance of different batches of predictors (“components of variation” in ANOVA terminology). As usual, we focus on estimation rather than testing: instead of testing the null hypothesis that a variance component is zero, we estimate the standard deviation of the corresponding batch of coefficients. If this standard deviation is estimated to be small, then the source of variation is minor—we do not worry about whether it is exactly zero. In the social science and public health examples that we focus on, it can be a useful research goal to identify important sources of variation, but it is rare that anything is truly zero.” (Gelman and Hill 2006, p. 490)

Batches of Parameters



Parameter Standard Deviations

Estos los estimamos con Nuestro modelo directamente (i.e. σ_L).



*“The **superpopulation** standard deviation, which represents the variation among the modeled probability distribution from which they were drawn, is relevant for determining the uncertainty about the value of a new group not in the original set of J.”*

*“The **finite-population** standard deviation of the particular J values of [some coefficient] describes variation within the existing data.”*

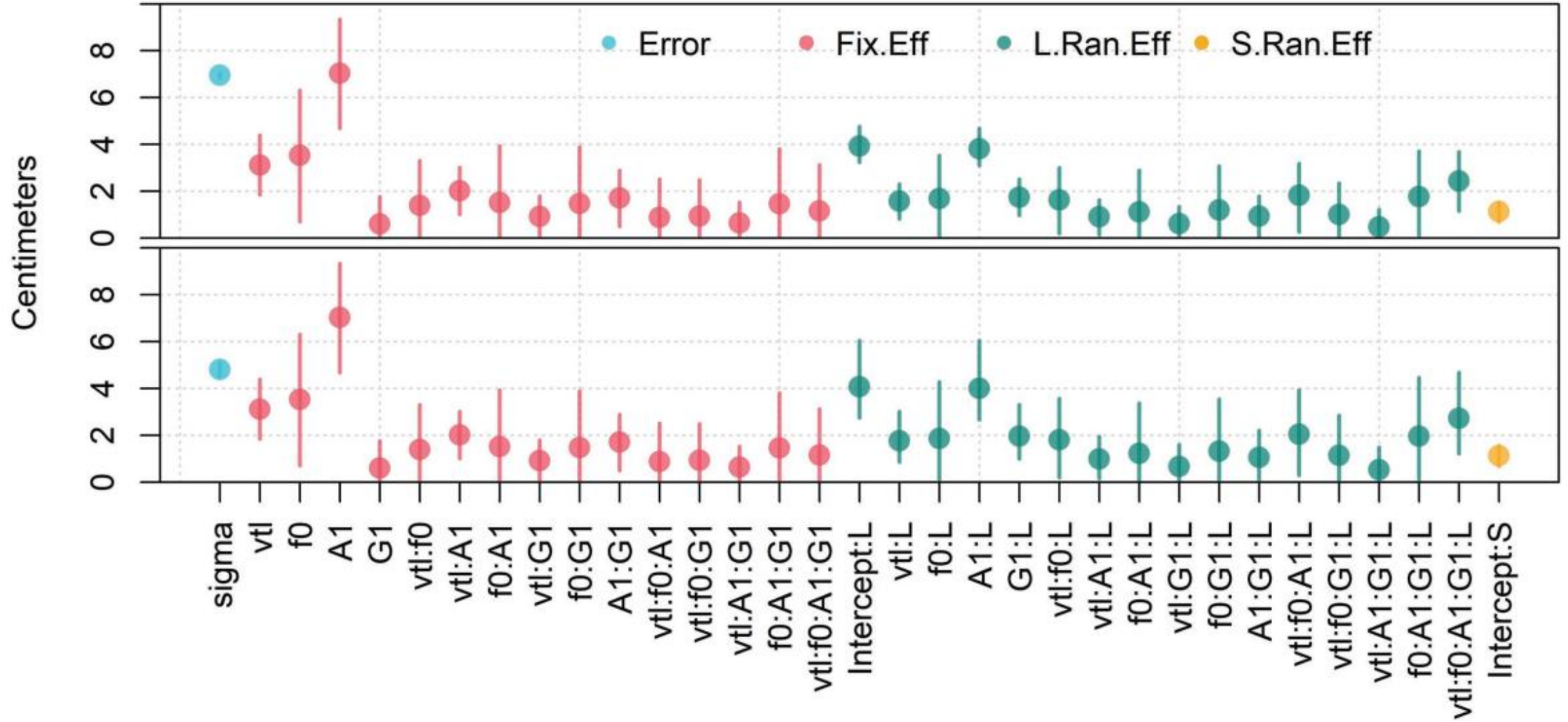


Esta es la desviación estándar real de los parámetros L (por ejemplo).

Bayesian ANOVA (BANOVA)

- 1 Fit the model with the structure you think is required to capture the variation in the data and answer your research questions.
- 2 Calculate the finite-population (and/or superpopulation) standard deviations for predictors or batches of predictors.
- 3 Make a plot comparing the magnitudes of different batches of predictors, and the uncertainty in the estimates. The authors refer to this as an **ANOVA plot**.
- 4 Use the ANOVA plot to make inferences about the relative importance of your predictors, and to guide your analysis.

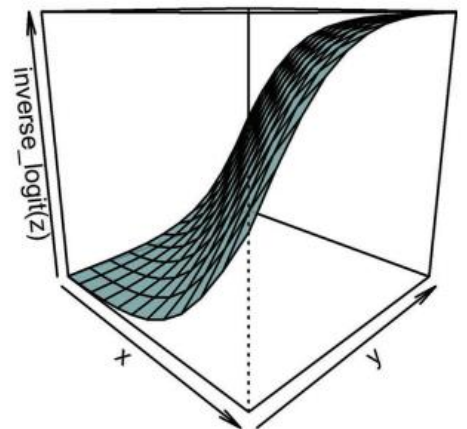
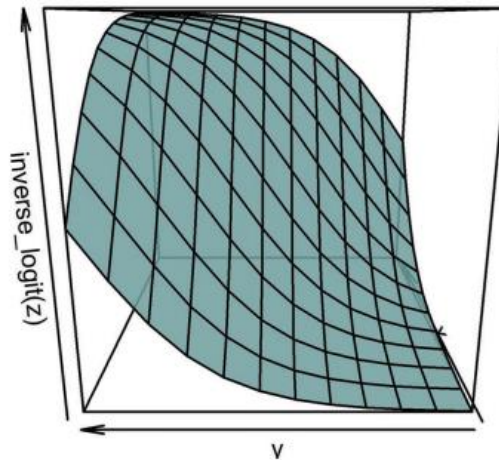
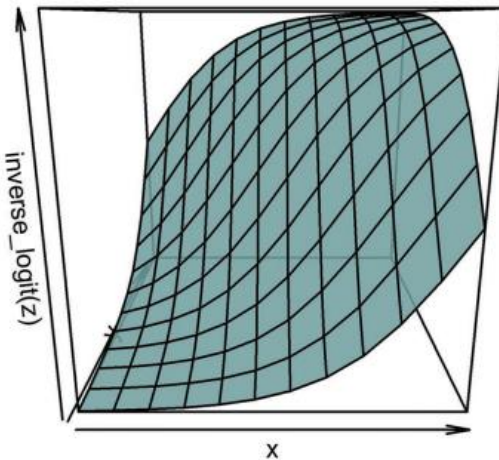
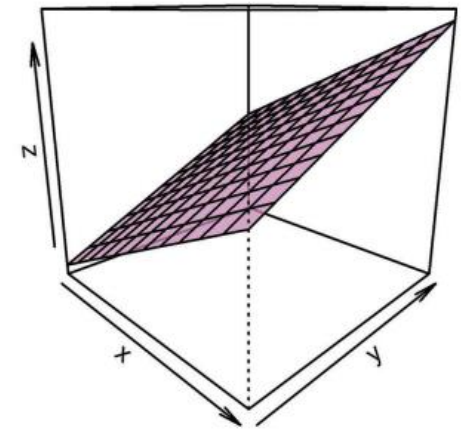
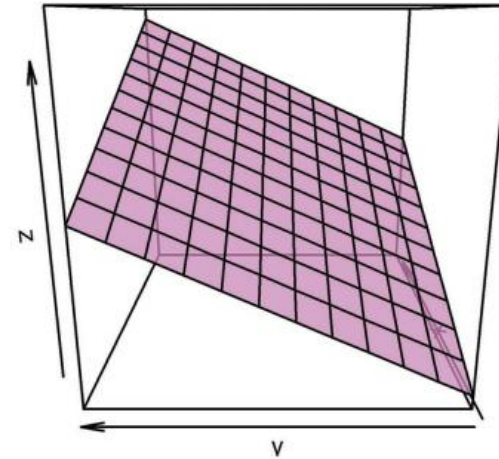
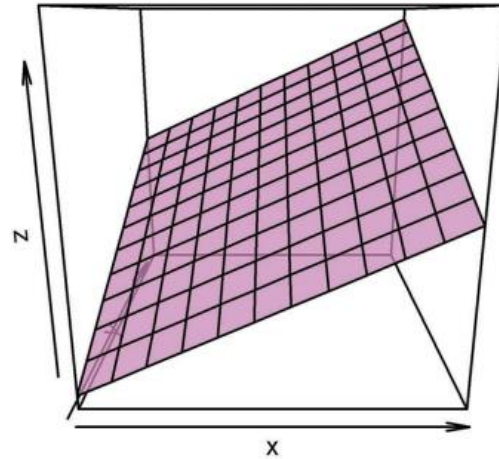
Bayesian ANOVA (BANOVA)



Logistic Regression with Multiple Quantitative Predictors

$$y \sim a + b \cdot x + c \cdot y$$

logits



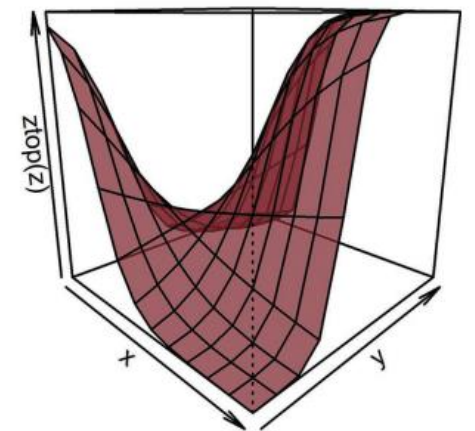
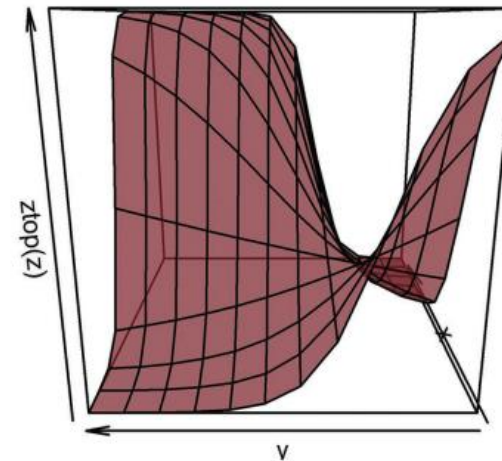
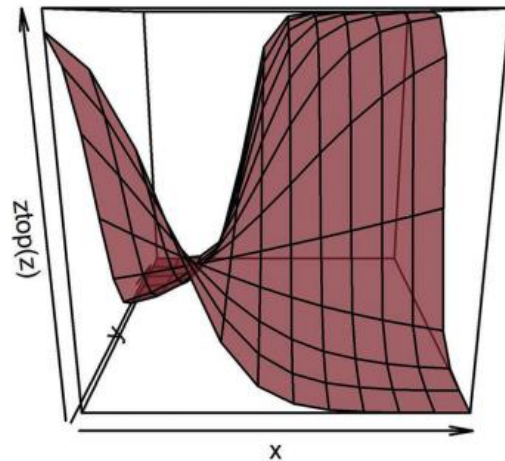
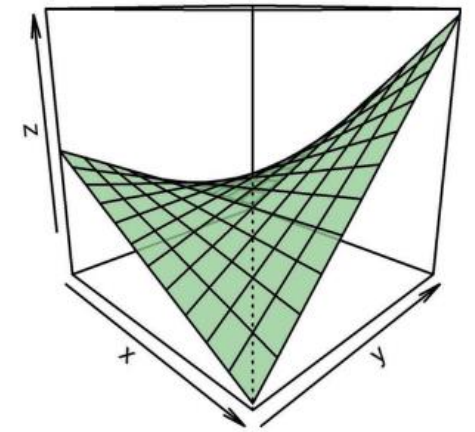
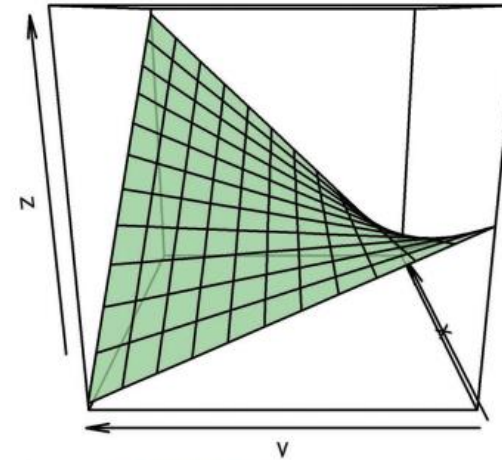
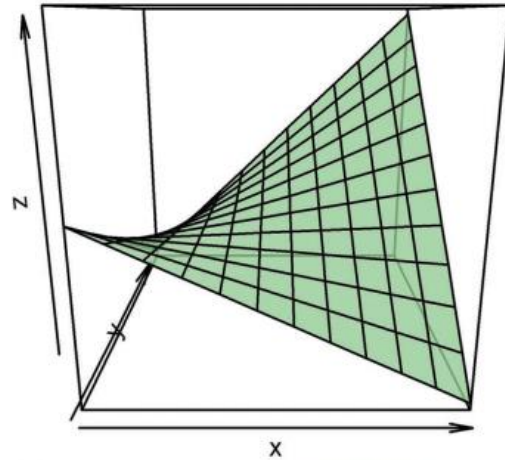
probabilidades



Logistic Regression with Multiple Quantitative Predictors

$$y \sim a + b \cdot x + c \cdot y + d \cdot x \cdot y$$

logits



probabilidades



Data and Research Questions

```
library (brms)
library (bmmmb)
options (contrasts = c('contr.sum', 'contr.sum'))

data (exp_data)

# create our dependent variable
exp_data$Female = ifelse(exp_data$G == 'f', 1, 0)

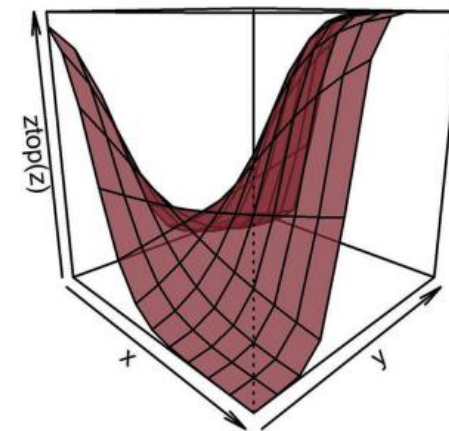
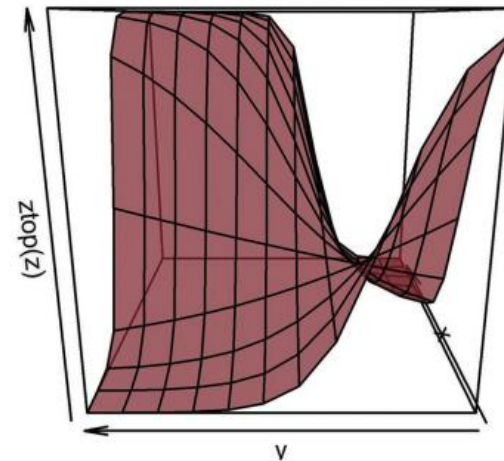
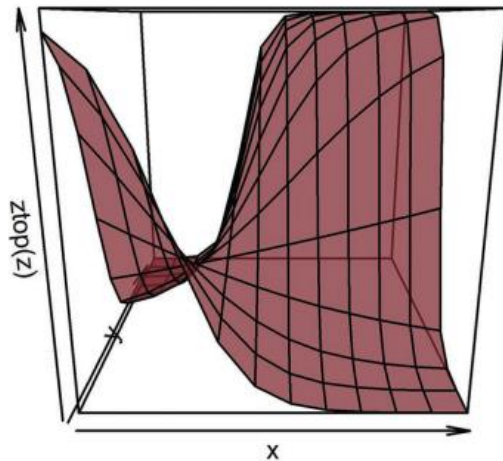
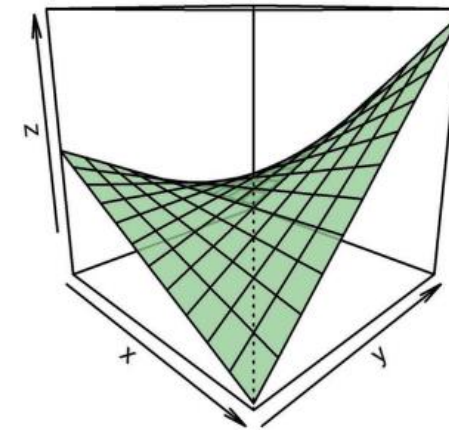
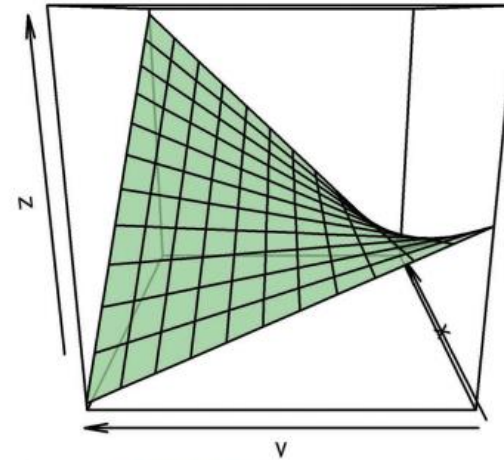
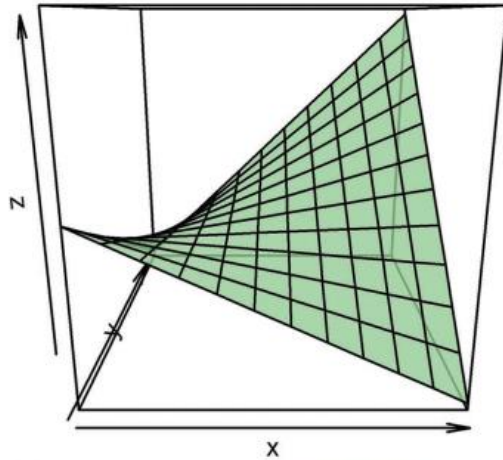
# center vtl
exp_data$vtl_original = exp_data$vtl
exp_data$vtl = exp_data$vtl - mean (exp_data$vtl)

# center and scale f0
exp_data$f0_original = exp_data$f0
exp_data$f0 = exp_data$f0 - mean(exp_data$f0)
exp_data$f0 = exp_data$f0 / 100
```

- (Q1) Can we use what we've learned so far to fit and evaluate a logistic regression model with two quantitative predictors?
- (Q2) Can we extend the concepts from Chapter 10 to classify speakers along two stimulus dimensions (f0 and VTL) using our logistic models?

Description of the Model

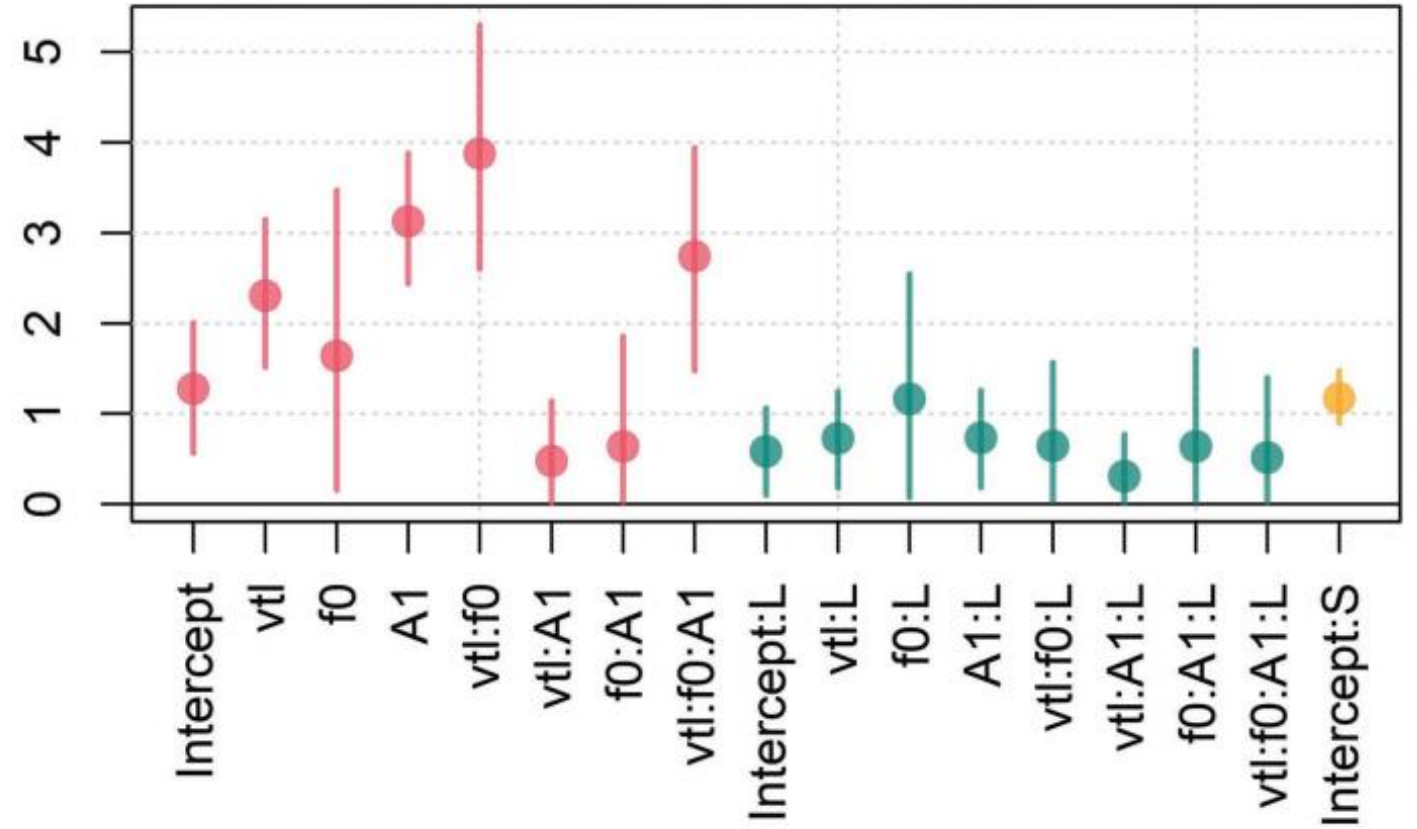
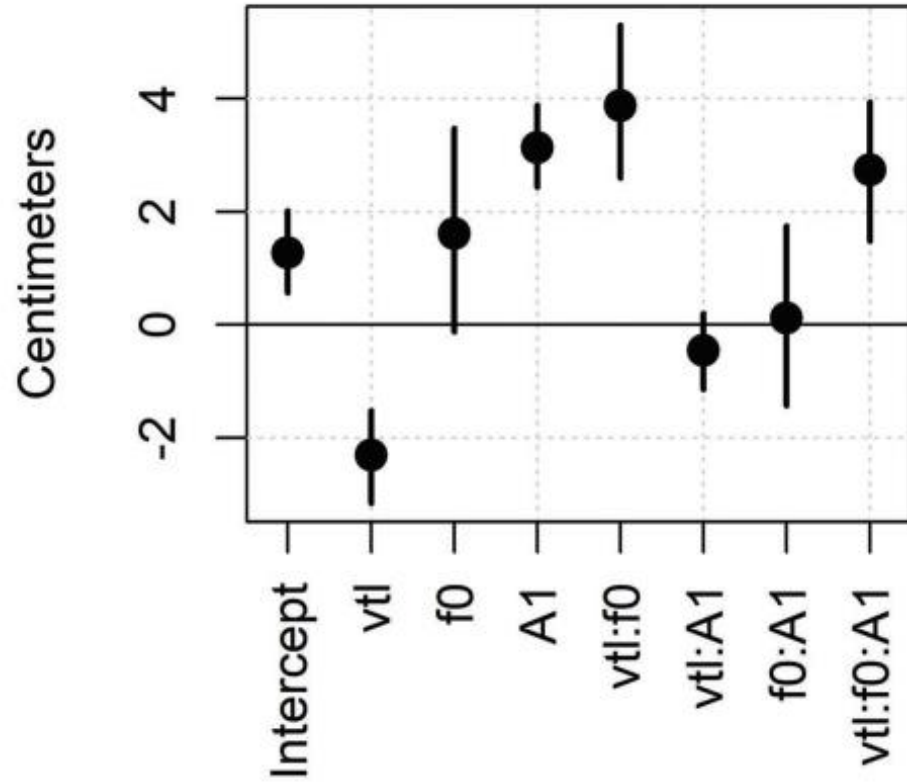
$$\text{Female} \sim \text{vtl} * \text{f0} * A + (\text{vtl} * \text{f0} * A | L) + (1 | S)$$



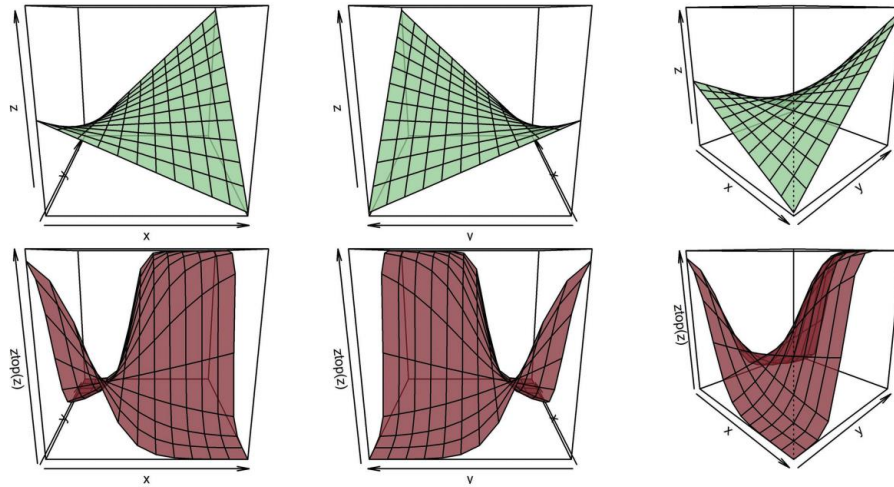
Fitting the Model

```
# Fit the model yourself
model_gender_vtl_f0 =
  brm (Female ~ vtl*f0*A + (vtl*f0*A|L) + (1|S), data=exp_data,
      chains=4, cores=4, family="bernoulli",
      warmup=1000, iter = 5000, thin = 4,
      prior = c(set_prior("student_t(3, 0, 3)", class = "Intercept"),
                set_prior("student_t(3, 0, 3)", class = "b"),
                set_prior("student_t(3, 0, 3)", class = "sd"),
                set_prior("lkj_corr_cholesky (2)", class = "cor"))))
```

Fixed Effects and BANOVA Plot

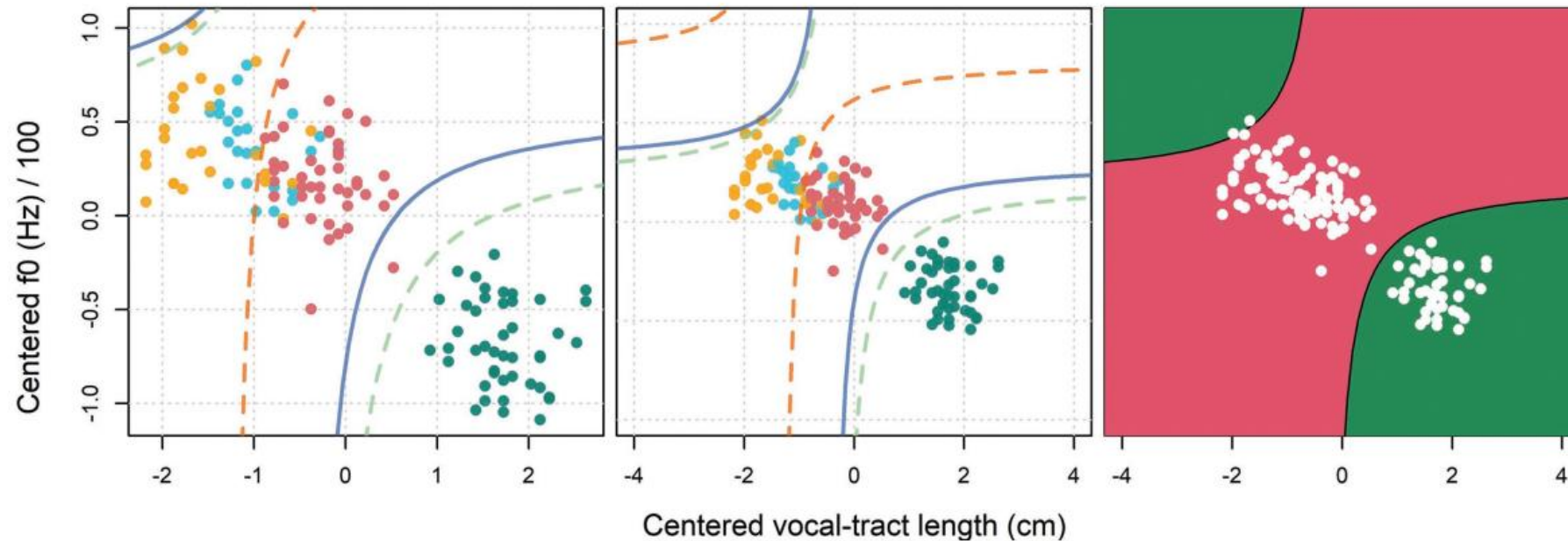


Categorization in Two Dimensions



$$0 = a + (b \cdot x) + (c \cdot y) + (d \cdot x \cdot y)$$

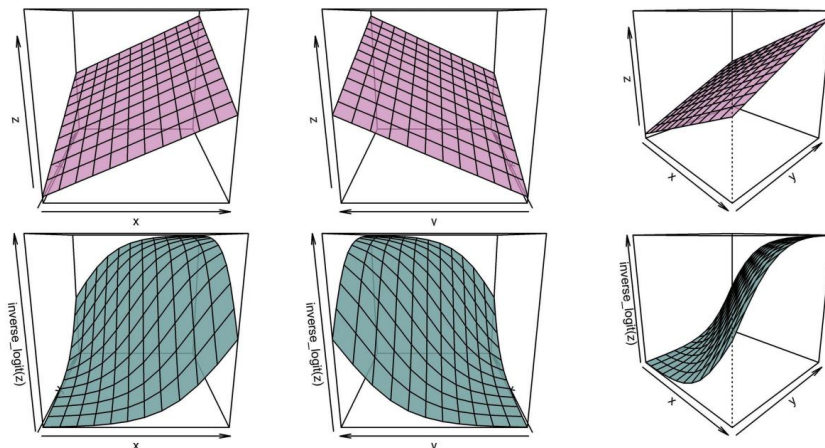
$$y = (-b \cdot x - a) / (d \cdot x + c)$$



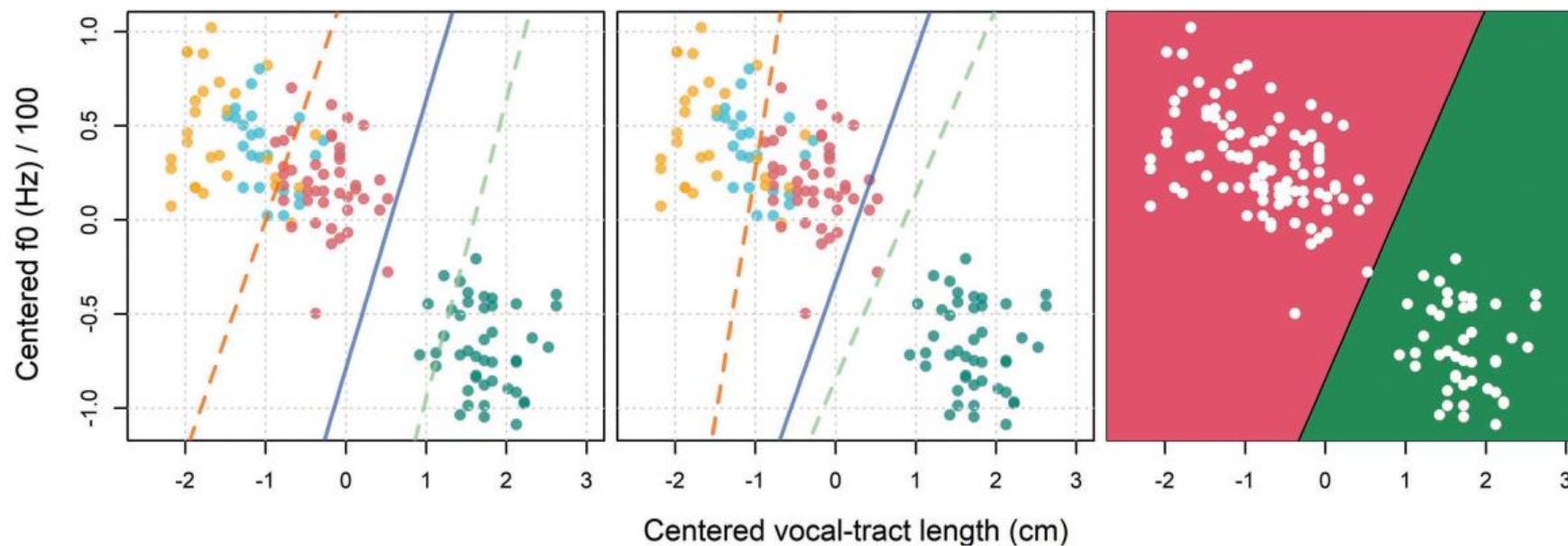
A Simpler Model

```
# Fit the model yourself
model_gender_vt1_f0_reduced =
  brm (Female ~ (vt1+f0)*A + ((vt1+f0)*A|L) + (1|S), data=exp_data,
      chains=4, cores=4, family="bernoulli",
      warmup=1000, iter = 5000, thin = 4,
      prior = c(set_prior("student_t(3, 0, 3)", class = "Intercept"),
                set_prior("student_t(3, 0, 3)", class = "b"),
                set_prior("student_t(3, 0, 3)", class = "sd"),
                set_prior("lkj_corr_cholesky (2)", class = "cor")))
```


Categorization in Two Dimensions



$$0 = a + (b \cdot x) + (c \cdot y)$$
$$y = -(b \cdot x - a) / c$$



Model Misspecification

```
brms::loo_compare (model_gender_vt1_f0, model_gender_vt1_f0_
reduced)
```

	<i>elpd_diff</i>	<i>se_diff</i>
## model_gender_vt1_f0	0.0	0.0
## model_gender_vt1_f0_reduced	-18.2	9.9

```
## Warning: Found 205 observations with a pareto_k > 0.7 in model
## 'model_gender_vt1_f0'. It is recommended to set 'moment_match = TRUE'
## in order to perform moment matching for problematic observations.
```

Pareto k Values

- Un diagnóstico del método utilizado para estimar LOO ELPD, que mide lo "inesperado" que es una observación.
- Valores de $\hat{k}$ menos de 0.5 son 'buenos', valores entre 0.5 y 0.7 son 'OK' (no tan buenos, pero tampoco malos), valores superiores a 0.7 son 'malos' y los valores superiores a 1.0 son 'muy malos'.
- Valores altos de $\hat{k}$ indican una gran discrepancia entre los datos y el modelo.

Model Misspecification

- 1 p_{loo} is *smaller* than the number of estimated parameters: If the estimated number of parameters (p) is relatively large relative to the number of observations (n), e.g. $p > n/5$, then the model may be too flexible, or your priors may be too weak.
- 2 p_{loo} is *much smaller* than the number of estimated parameters: The model is likely to be **misspecified**. A misspecified model is one whose structure contains important differences compared to the processes being modeled.
- 3 p_{loo} is *greater than* the number of parameters: The model is likely to be ‘*badly*’ misspecified.

```
# actual number of estimated parameters
ncol (bmmmb::get_samples(model_gender_vtl_f0)) - 2
## [1] 304

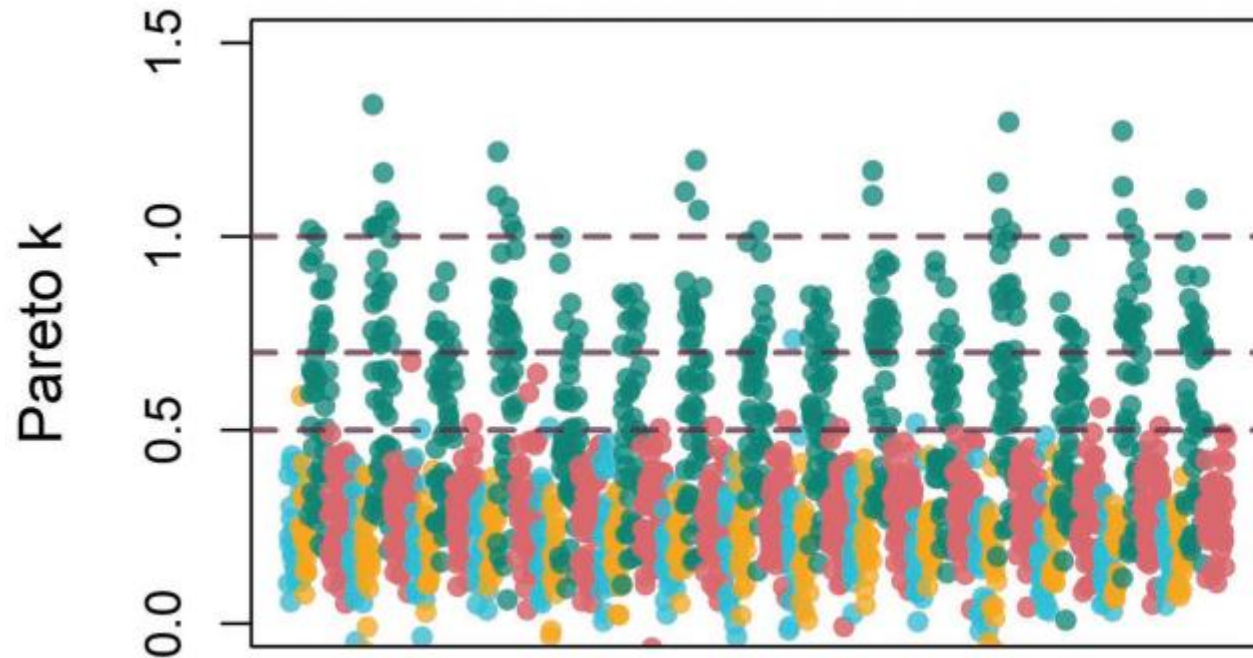
# number of observations
nrow (model_gender_vtl_f0$data)
## [1] 2085

# information related to loo criterion
model_gender_vtl_f0$criteria$loo$estimates
##           Estimate      SE
## elpd_loo    -541.5  23.405
## p_loo        105.8   6.637
## looic        1083.0  46.810
```

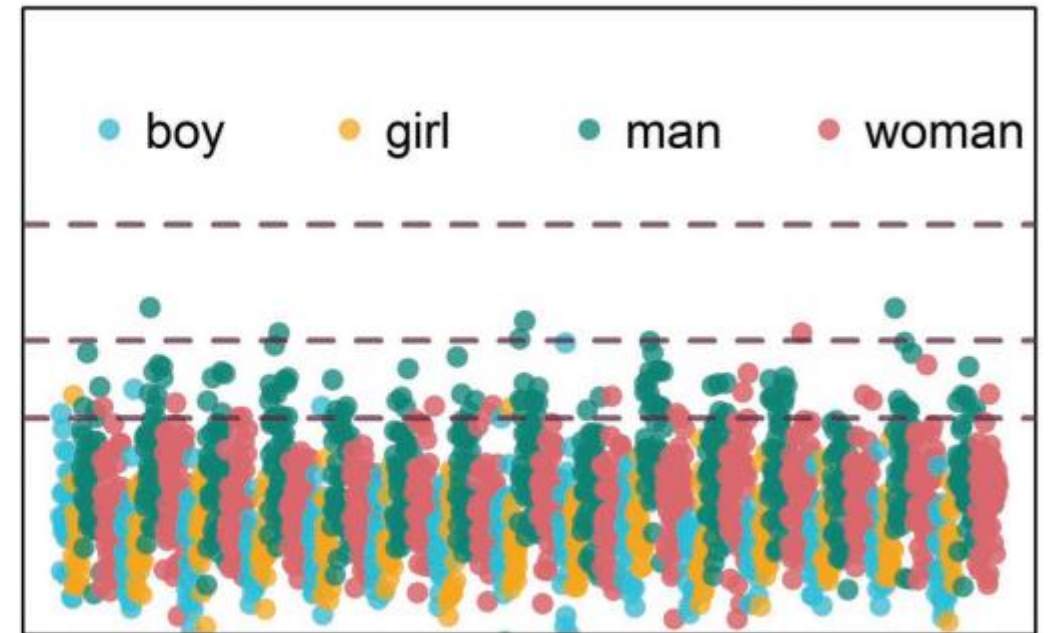
Pareto k Values

```
pareto_k = model_gender_vtl_f0$criteria$loo$diagnostics$pareto_k
```

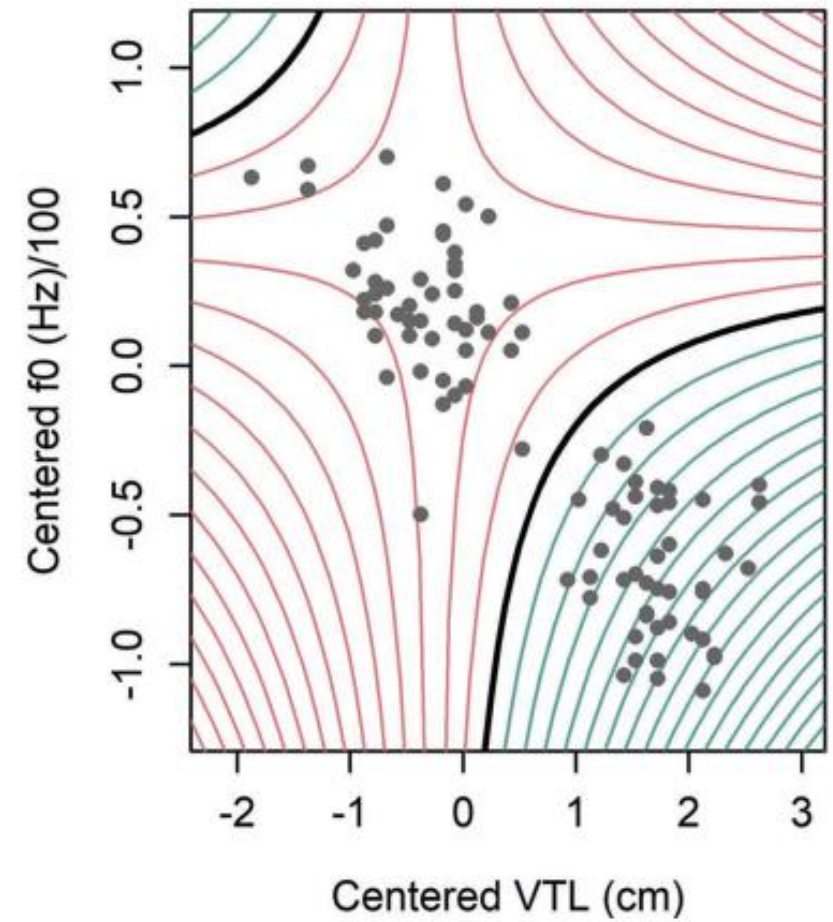
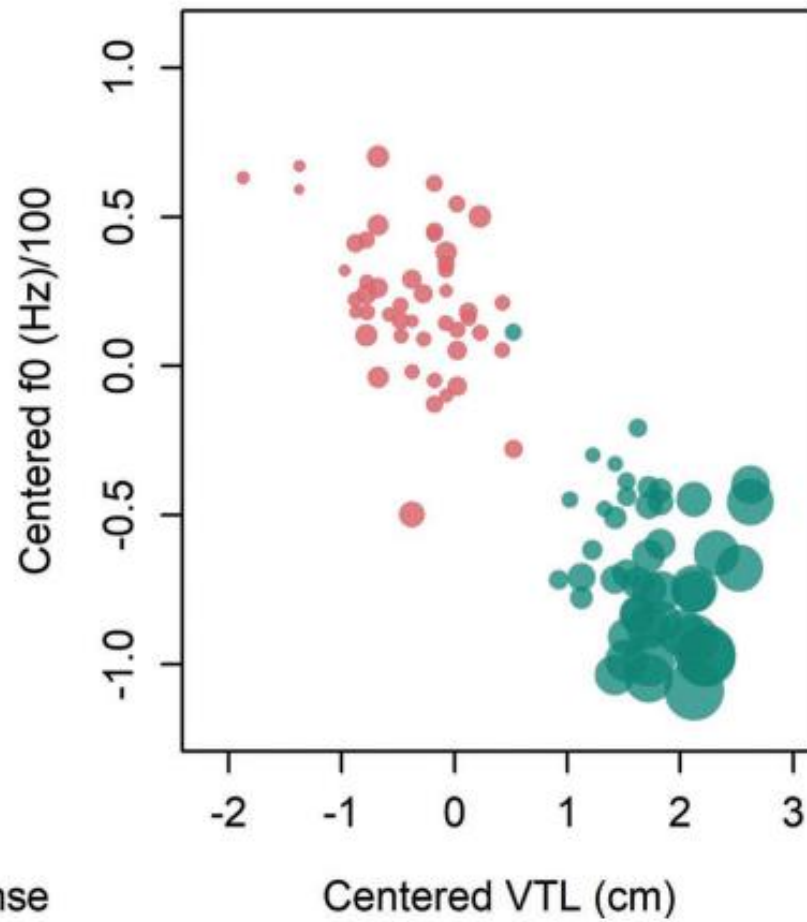
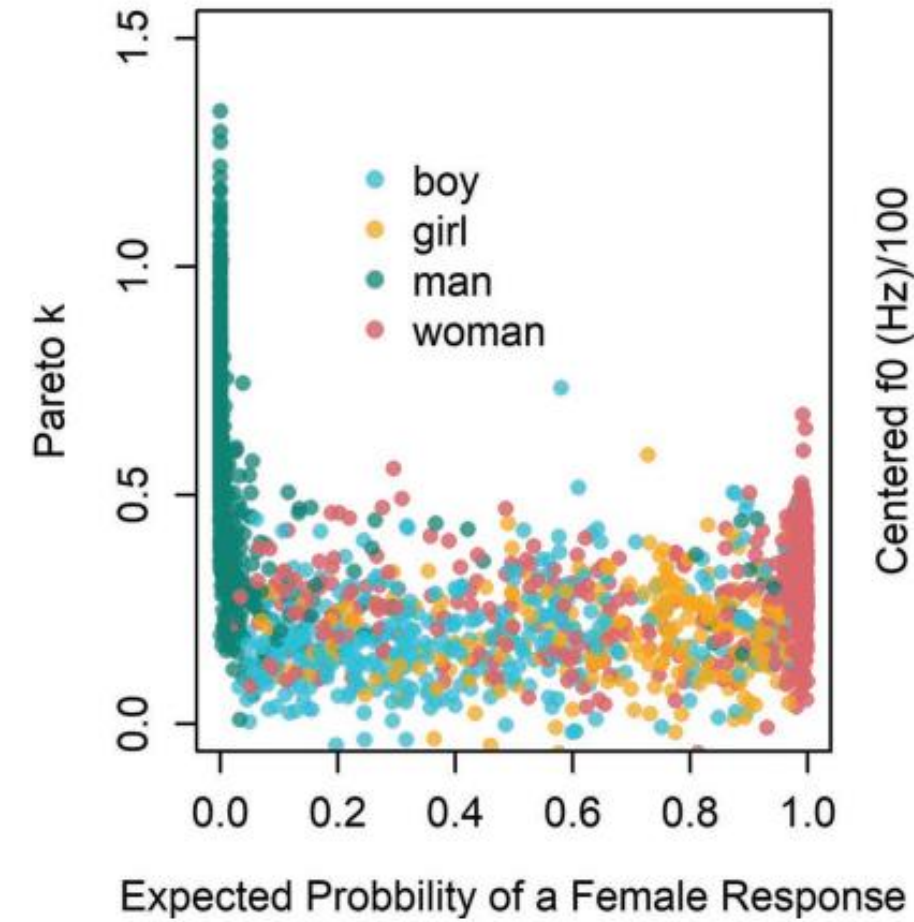
Original Model



Reduced Model

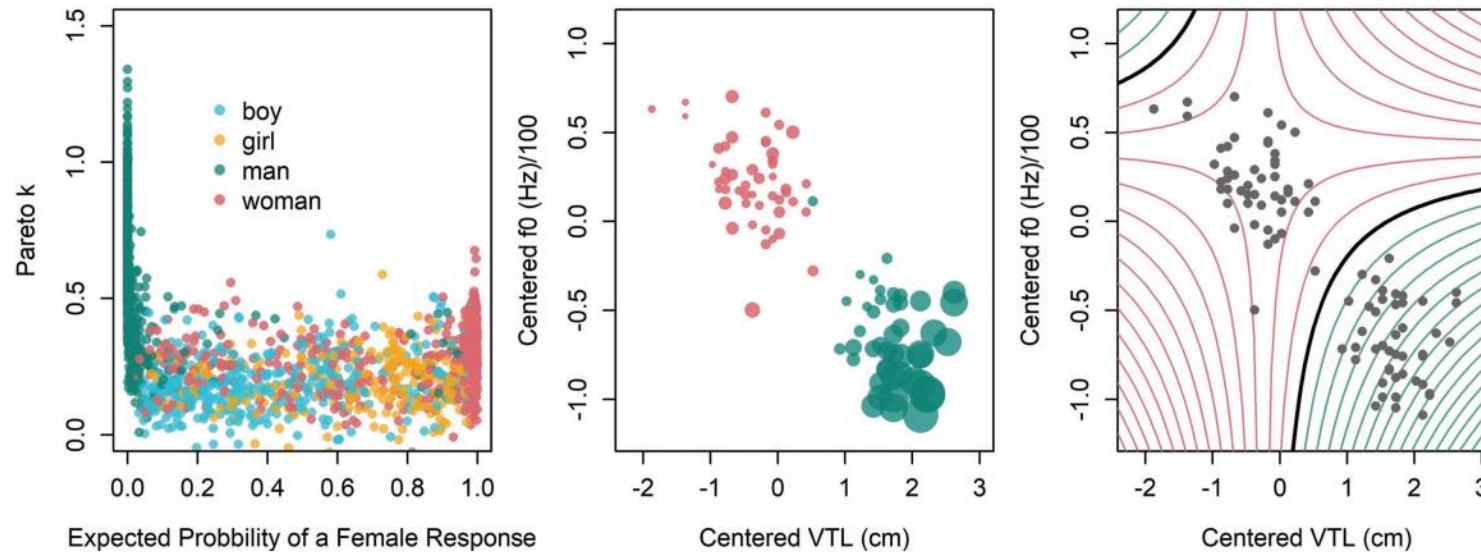


Pareto k Values



Model Selection: Statistical

- Nuestro modelo complicado con priores débiles se comportó mal. Los problemas solo se hicieron evidentes cuando se agregó el criterio LOO.
- Esta es una buena manera de diagnosticar problemas "invisibles" con sus modelos.



Model Selection: Common Sense

- Es imposible que los hablantes con el f_0 más alto y los trectos vocales más cortos sean percibidos como hombres adultos.
 - Es razón suficiente para desconfiar de este modelo.
- El modelo más simple se alinea con nuestro conocimiento del dominio y las expectativas previas, y eso por sí solo es una buena razón para preferirlo.

